

/Users/halpern/teaching/352/labs/manual/master/rev\_2022/alpha.tex

# Range of Alpha Particles in Air

## 1 Purpose

1. To study the energy loss of charged particles in matter. and to measure the energy loss as a function of particle energy.

## 2 Theory

As charged particles traverse matter, they lose energy by ionizing atoms along their path. As the amount of energy lost per collision is approximately constant, each particle will have a path length or range in a given stopping material dependent on its incident energy. Thus for any particular type of particle the range is a definite function of the energy as shown in Figure ???. The mean range of particles is defined as shown in Figure ??, where  $n_o$  is the number of particles reaching a detector with no stopping material present and  $n$  is the number reaching the detector with stopping material of  $x$  cm between the source and the detector.

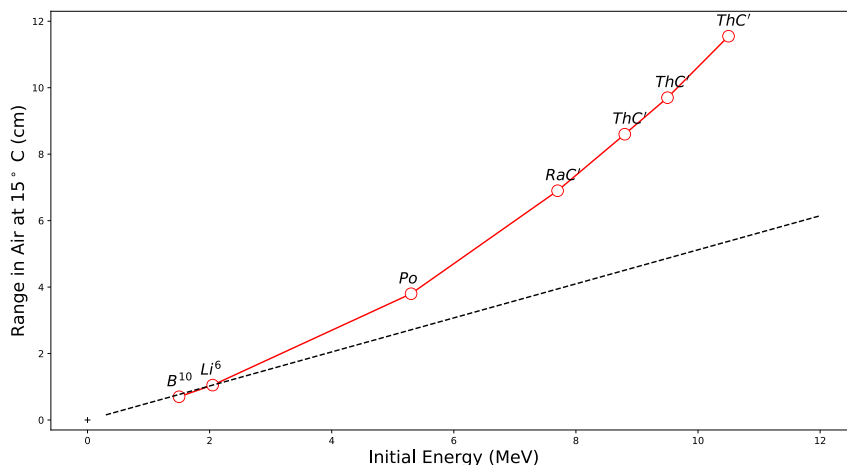


Figure 1: Range-energy relationship for  $\alpha$  particles.

The mean range of alpha particles can also be found by plotting the differential range or the difference in the number of alpha particles detected for each increment of absorber as shown in Figure ???. The distribution of ranges about the mean range is Gaussian i.e.

$$\frac{\Delta n_i}{n_0} = \frac{\Delta x_i}{\alpha\sqrt{\pi}} e^{-\frac{(x_i - R)^2}{\alpha^2}} \quad (1)$$

where  $\Delta n_i/n_0$  is the fraction of particles having range between  $x_i$  and  $x_i + \Delta x_i$ . The range straggling parameter,  $\alpha$ , is the half width of the range distribution at  $1/e$  of maximum as shown in Figure ???. The best measurements have shown that  $\alpha = 0.15R$

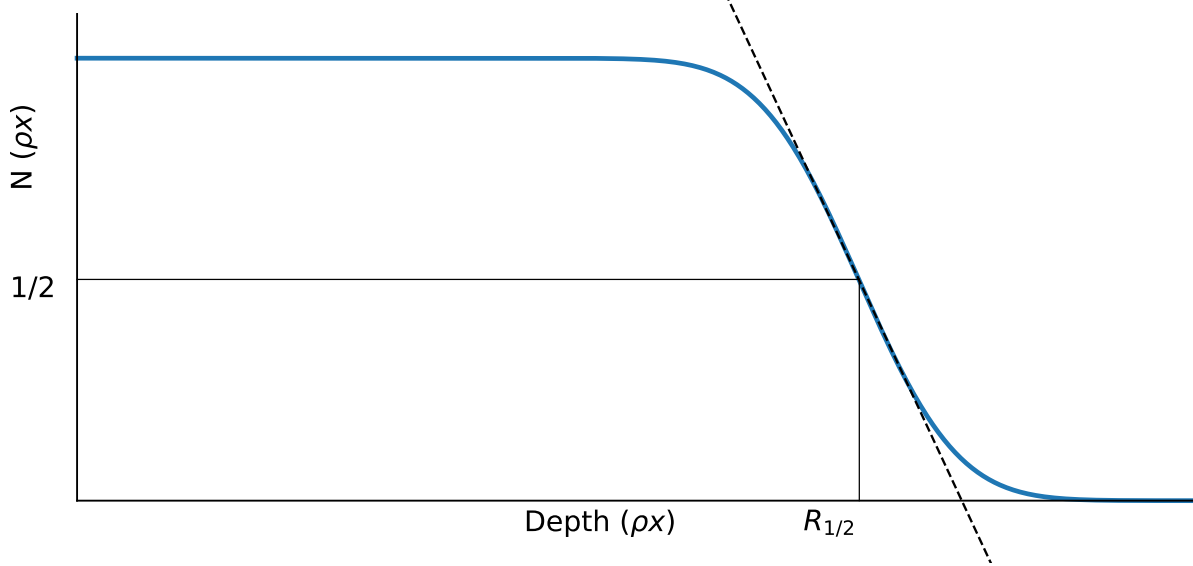


Figure 2: The extrapolated number-distance range  $R_n$  exceeds the mean range  $R$  by  $0.886\alpha$ , where  $\alpha$  is the range-straggling parameter.

The distribution is broadened by alpha particles scattering off the inside of the chamber, the finite energy resolution of the detector, and the thickness of the source itself. If the width of the differential range curve is broad or asymmetric, the energy might be best determined using the extrapolated range.

The rate of energy loss of alpha particles is given as

$$-\frac{dE}{dx} = \frac{4\pi Z^2}{m_e v^2} \left( \frac{e^2}{4\pi\epsilon_0} \right)^2 n_e \ln \frac{2m_e v^2}{I} \quad (2)$$

where  $I$  is some average ionization potential of the material,  $Z$  is the atomic number of the incoming particle,  $v$  is its (non-relativistic) velocity, and  $n_e$  is the electron density of the scattering material. This is plotted for Nitrogen gas in Figure ???. Notice that the stopping power drops at high particle energy with a rate between  $1/\text{Energy}$  and  $1/\text{Velocity}$ .

While the concepts explored in this experiment have plenty of applications in nuclear physics and nuclear medicine, there are less obvious applications. In particular, semiconductors are

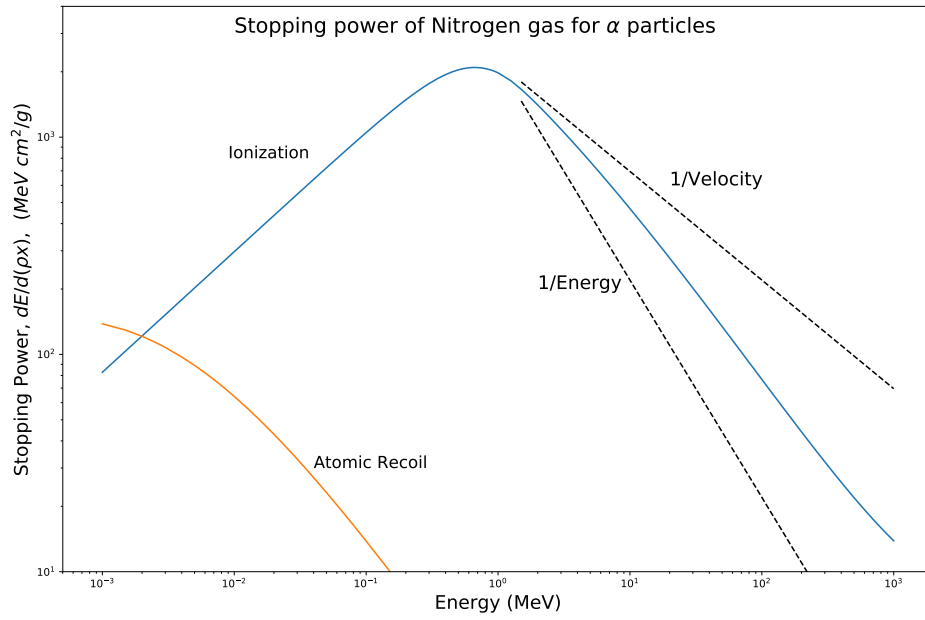


Figure 3: The stopping power of Nitrogen gas, from NIST (<https://physics.nist.gov/PhysRefData/Star/Text/ASTAR.html>). At low energies the  $\alpha$  particle does not get near enough to the core of an atom to ionize it. At high energies it goes by too quickly for the electrical forces to have a large effect.

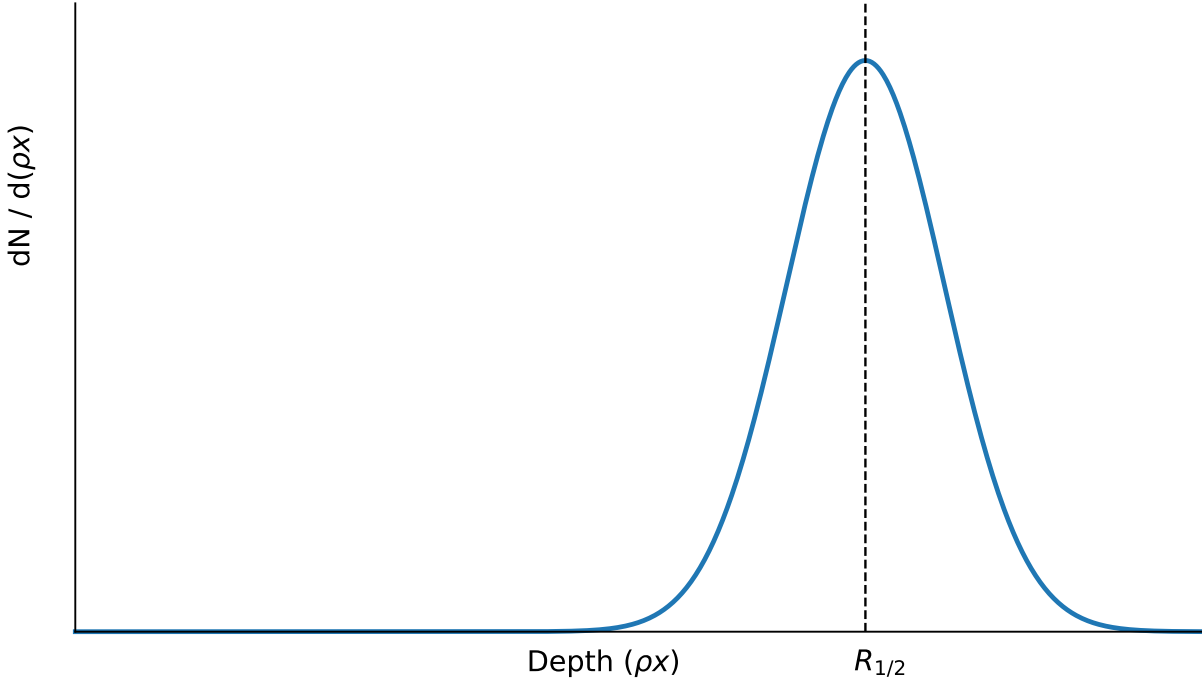


Figure 4: Schematic “number-range curve,” illustrating a symmetric distribution of ranges about the mean range  $R$ . The “range straggling parameter”  $\alpha$  is  $\sqrt{2}$  times the standard deviation of the range distribution.

often doped in a similar way, although the dopants are normally neutral atoms and are slowed by collisions with the lattice.

## 2.1 Chamber Air Pressure Measurement

A solid state pressure transducer is used to convert pressure into voltage readings. The transducer uses strain gauges in a bridge configuration to measure the displacement produced in a solid plate by having an absolute vacuum (static) applied to one side and the working pressure applied to the other.

The transducer is calibrated at  $P = 1$  atm and vacuum. Measure the atmospheric pressure when you do the lab using the Fortin barometer on the end of the bookshelves near the lab door. The other calibration point is obtained by pumping the chamber to  $P \approx 0$  mmHg using the vacuum pump. A linear interpolation is used to determine the chamber pressure from the transducer response.

In this experiment, the range will be varied by increasing the air pressure in the vacuum chamber. Adjust the source-detector distance to  $d_0 \approx 4.5$  cm, so that the effective distance,

$d$ , between the source and detector will be approximately given by

$$d = \frac{P}{P_{atm}} d_0 \quad (3)$$

### 3 Procedure

**BEFORE STARTING:** Read the precautions on using the surface barrier detector located at the lab bench. Do not touch either the detector or the radioactive sample of ( $\sim 1\mu\text{Ci } ^{241}\text{AmO}_2$ ).

1. Connect the apparatus as shown in Figure ??, and turn on the equipment. Open the UC530 application. Ensure that the bias voltage (“high voltage”) for the detector is set to zero. Check that when you turn it on it will be negative (hardware switch in one setup, software in the other).
2. Calibrate the pressure transducer as outlined above, and verify the separation between source and detector is  $\approx 4.5$  cm, recording the actual value. Evacuate the chamber when finished.
3. START THE NEGATIVE BIAS VOLTAGE AT 10-20 VOLTS AND TURN UP SLOWLY, MAKING SURE NOT TO EXCEED THE MAXIMUM ALLOWED FOR THE DETECTOR, THE BIAS POWER SUPPLY GOES UP TO 1000V!

| Detector serial number | Max voltage |
|------------------------|-------------|
| 16-625J                | -100 volts  |
| 16-721J                | -75 volts   |
| 16-469A                | -100 volts  |

4. Adjust the bias voltage for best signal to noise and adjust the amplifier gain. On one set-up with the older data acquisition system, adjust the gain so that the pulses are approximately 8 volts on the oscilloscope. With the newer data acquisition system you should set the gain so that the highest energy pulses are on scale.
5. Observe the pulse height spectrum using the pulse height analyzer, perhaps re-adjusting the amplifier gain so that the energy spectrum is at the right of the display. Note: to clear the spectrum in the older setup, use control-F2.
6. Close the valve to the pump and observe the change in pulse height as air is admitted to the chamber.
7. Now re-evacuate the chamber and record the count rate and the centroid on the pulse height analyzer. You will want to set your Region-Of-Interest (ROI) to be just above the noise in the first several channels. Observe the rate and centroid channel at several

different pressures. The points near the end of the range (near atmospheric pressure) are the most important, so get plenty of data in this range. How would you calculate the error on the number of counts?

8. Convert the transducer voltage readings into pressure values using your calibration.
9. Convert the channel of the centroid into energy assuming a linear calibration. Take the centroid channel at vacuum to equal the full source energy and assume channel 0 is at 0 MeV.
10. Plot the centroid of the energy as a function of effective distance in air and fit a curve to the data (the curve you fit should be based on Eq. ??).
11. Plot the derivative of the curve, again as a function of effective distance, and display together with the curve from above on the same axis. This gives you the energy loss per unit length and is called the Bragg Curve. Is it what you expected? Actually, your data include the full energy distribution at each pressure. Is there something you can do with that to obtain a more precise Bragg curve?
12. Plot the count rate as a function of effective distance. Also plot the derivative of the curve as a function of effective distance and again display the curves together.
13. Determine the range and range straggling parameter.

When you are finished with the equipment, please turn the bias power supply back down to 0 V, and vent the pump when you turn it off: fully open the two air admittance valves and then open the pump valve and then shut off the pump.

## 4 References

- Evans, **The Atomic Nucleus**, Pages 650 - 667.
- Melissinos, **Experiments in Modern Physics**, sections 5.2.2, 5.2.3, and 5.5.3 (first edition).